

1.2.2 Equations of Motion

The double-pendulum system, along with the angles and rods lengths notations, are shown in Figure 1.

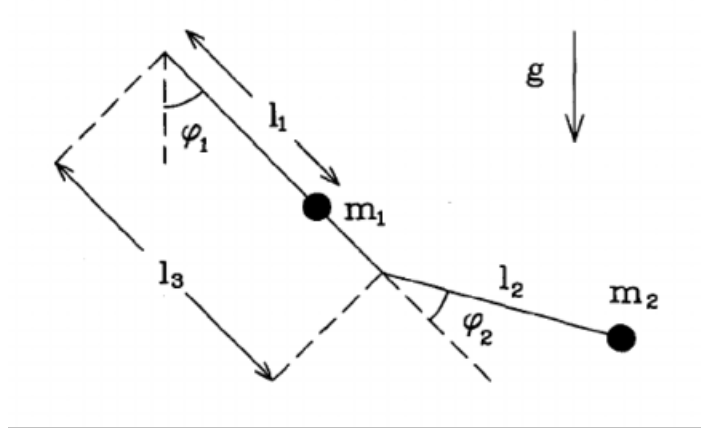


Figure 1: Schematic diagram of the double-pendulum system. The system consists of two masses m_1 and m_2 , connected by rigid rods of lengths l_1 and l_2 , respectively. The angles φ_1 and φ_2 represent the angular displacements of the two pendulums. This diagram was taken from [1] (Fig.1).

The dynamics of the double pendulum are governed by the Euler-Lagrange equations, which are derived from the Lagrangian of the system:

$$\mathcal{L} = T - V, \quad (1)$$

where T and V are the kinetic and potential energy, respectively.

The kinetic energy of the system accounts for the rotational and translational motion of the two masses, m_1 and m_2 :

$$T = \frac{1}{2} (I_1 + m_2 l_3^2) \dot{\varphi}_1^2 + \frac{1}{2} I_2 (\dot{\varphi}_1 + \dot{\varphi}_2)^2 + m_2 l_2 l_3 \dot{\varphi}_1 (\dot{\varphi}_1 + \dot{\varphi}_2) \cos(\varphi_2), \quad (2)$$

where l_2 and l_3 are the lengths of the pendulum arms, $\dot{\varphi}_i$ are the angular velocities, and I_i are the moments of inertia or the arms, computed by measuring each pendulum time period T_i , such as:

$$I_i = m_i g l_i \left(\frac{T_i}{2\pi} \right)^2, \quad i = 1, 2.$$

This is done experimentally by measuring the masses (using a scale), the arm lengths (using a ruler), and the time period by attaching separately each arm to the main rotational axis of the system and measure the time it takes for each of them to complete several (chosen to be 20, to reduce the error accordingly) full swings (periods).

The potential energy of the system is due to gravity acting on the two masses:

$$V = g (m_1 l_1 + m_2 l_3) (1 - \cos(\varphi_1)) + m_2 g l_2 [1 - \cos(\varphi_1 + \varphi_2)]. \quad (3)$$

The equations of motion (denoted by EOMs, from here) are given by the Euler-Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\varphi}_i} \right) - \frac{\partial \mathcal{L}}{\partial \varphi_i} = 0, \quad i = 1, 2, \quad (4)$$

where φ_i are the angular displacements of the first and second pendulums, respectively, as defined in Figure 1. Substituting the expressions for T and V into the Lagrangian and applying the Euler-Lagrange equations yields a set of coupled nonlinear differential equations describing the motion of the double pendulum.

1.2.3 Small-Angles Approximation

For small angles φ_1 and φ_2 , the EOMs become linear and become:

$$\begin{pmatrix} \ddot{\varphi}_1 \\ \ddot{\varphi}_2 \end{pmatrix} = \frac{1}{d} \begin{pmatrix} c_1 & c_2 \\ c_3 & c_4 \end{pmatrix} \begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix}, \quad (5)$$

where the coefficients are defined as follows:

$$c_1 = m_2 g l_2^2 l_3 - I_2 g (m_1 l_1 + m_2 l_3), \quad (6)$$

$$c_2 = m_2^2 g l_2^2 l_3, \quad (7)$$

$$c_3 = I_2 g (m_1 l_1 + m_2 l_3) + m_2 g l_2 (m_1 l_1 l_3 - I_1 - m_2 l_2 l_3), \quad (8)$$

$$c_4 = -m_2 g l_2 (m_2 l_3 + m_2 l_2 l_3^2 + I_1), \quad (9)$$

$$d = I_2 (I_1 + m_2 l_3^2) - m_2^2 l_2^2 l_3^2. \quad (10)$$

The received EOMs are integrable, and represent well-known harmonic oscillators. There are two normal modes, and their frequencies are given by:

$$\omega_{1,2}^2 = \frac{-(c_1 + c_4) \pm \sqrt{(c_1 + c_4)^2 - 4(c_1 c_4 - c_2 c_3)}}{2d}. \quad (11)$$

1.2.4 Fourier Analysis for Small Angles Approximation

To validate the theoretical frequencies, we will measure the pendulum angles for small initial angle, and perform a discrete Fourier transform (DFT) to observe the experimental spectrum. The DFT of the angular displacement signals is computed using:

$$P_\varphi(f) = |\mathcal{F}[\varphi(t)]|^2, \quad (12)$$

where $P_\varphi(f)$ is the power spectral density, and $\mathcal{F}[\cdot]$ represents the Fourier transform:

$$\mathcal{F}[\varphi(t)] = \int_{-\infty}^{\infty} \varphi(t) e^{-i2\pi f t} dt. \quad (13)$$

We use the MATLAB `fft` function to compute the DFT efficiently, which uses the Fast Fourier Transform (FFT) algorithm. The algorithm transforms the time-domain signal, $\varphi(t)$, into its frequency-domain representation by decomposing it into a sum of sinusoidal components. The output includes both the amplitudes and phases of these components. The sampling frequency f_s is determined from the mean time step of the data:

$$f_s = \frac{1}{\Delta t}, \quad \Delta t = \text{mean}(\Delta t_i), \quad (14)$$

where Δt_i is the time interval between successive measurements. Peaks in the power spectral density correspond to the dominant frequencies of the system. These frequencies, $f_{\text{exp},1}$ and $f_{\text{exp},2}$, are compared to the theoretical frequencies $f_{\text{theo},1}$ and $f_{\text{theo},2}$, where $f = \omega/(2\pi)$, calculated from the linearized equations of motion for small angles - Equation (11).

1.3 Lyapunov Exponents and Large-Angle Motion

For large-angle motion in the double pendulum, the dynamics become non-integrable and sometimes chaotic (only above a threshold we will examine). A key characteristic of chaos is the divergence of nearby trajectories in phase space, which can be quantified using Lyapunov exponents. These exponents measure the average exponential rate at which initially close trajectories will separate over time. If we consider two trajectories in phase space, $\varphi_i^I(t)$ and $\varphi_i^{II}(t)$, starting with a small initial difference $d_0 \equiv \delta\varphi_i(0)$, their separation grows as:

$$d \equiv |\varphi_i^{II}(t) - \varphi_i^I(t)| = d_0 e^{\lambda t} \propto e^{\lambda_1 t}, \quad (15)$$

or equivalently:

$$\log(d) \equiv \log(d_0) + \lambda t, \quad (16)$$

where λ_1 is the largest Lyapunov exponent, indicating the rate of divergence. The Lyapunov exponent λ will therefore be experimentally extracted using the slope of a linear fit applied on initial growths in trajectory separations of experimental pairs (each pair aims to start the experiment with an identical initial condition). In an n -dimensional phase space, there are n Lyapunov exponents, $\lambda_1, \lambda_2, \dots, \lambda_n$. These describe the rate at which a small sphere of initial conditions evolves into an ellipsoid over time, and they are ordered such as:

- λ_1 is the largest Lyapunov exponent and corresponds to the most rapid direction of divergence.
- λ_n is the smallest and typically indicates contraction.

It is worth noting that the trajectory separation, in leading order, grows exponentially with the largest Lyapunov exponent. The sum of the Lyapunov exponents for a conservative system, such as the double pendulum, is zero, reflecting the conservation of phase-space volume (Liouville's theorem). A positive Lyapunov exponent indicates chaos in the system, as small uncertainties in initial conditions grow exponentially, making long-term predictions impossible. Conversely, a zero or negative Lyapunov exponent corresponds to periodic or stable behavior. The Lyapunov exponents can be calculated by solving the nonlinear EOMs numerically, Figure 2 (reproduced from Fig.3 in [1]) shows the numerical calculation of the most positive Lyapunov exponent for the double pendulum, with parameters appearing in Equation (25) in [1], under the initial conditions:

$$\varphi_1(0) = -\varphi_2(0) = 120^\circ, \quad \dot{\varphi}_1(0) = \dot{\varphi}_2(0) = 0. \quad (17)$$

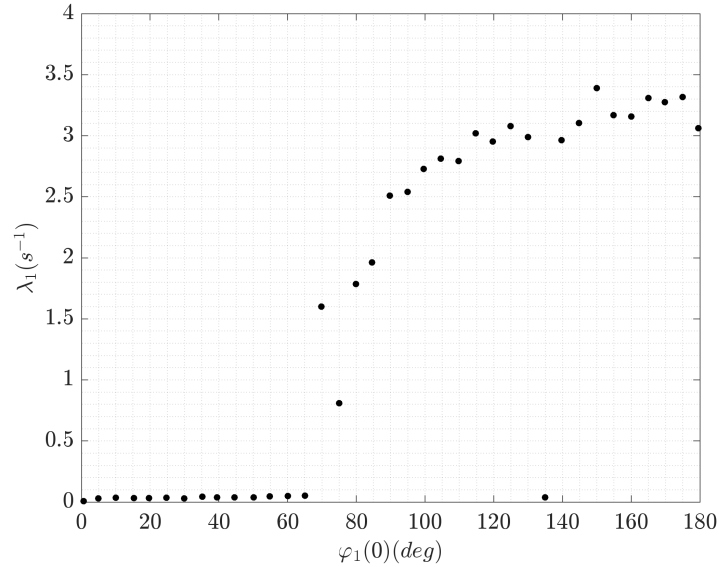


Figure 2: Reprint of Fig.3 in [1]. Numerical calculation of the most positive Lyapunov exponent for different initial conditions $\varphi_1(0)$. In all cases $\varphi_1(0) = -\varphi_2(0)$, and $\dot{\varphi}_1(0) = \dot{\varphi}_2(0) = 0$.

We can clearly see that, for this system's parameters, starting from approximately from $\varphi_1(0) \approx 70$ degrees, the most positive Lyapunov exponent grows significantly; exhibiting a transition to chaos. In the results section, we will reproduce this graph, for our system's parameters, because of its importance to characterize the interval of $\varphi_0(0)$ where transition to chaos occur. It is worth noting that the results in Figure 2 are for a different double-pendulum setup with its own mechanical properties, which also differs from ours. Therefore, we do not expect the transition to chaos to be at the same initial angle, although the general behavior should look similar.

1.4 Poincaré Maps

Poincaré maps provide a visual way to analyze whether a system is chaotic or periodic.

In order to create a Poincaré map, we examine the trajectory of the system in phase space and select a cross-sectional plane appropriate to our analysis. Each time the trajectory intersects this plane, a point is marked on the map, all collected to form a Poincaré map. This tool is particularly useful for understanding the dynamics of the system since it reduces its dimensions of the system, and allows us to reveal important system characteristics, like:

- **Phase-space periodicity:** If the system is periodic, its trajectory in phase space will repeat, forming a closed orbit. On the Poincaré map, this corresponds to recurring points at the same locations. In an ideal non-chaotic system without energy dissipation, these points will precisely overlap. However, in real systems with energy loss, the recurrence may exhibit slight spreading, creating a dense closed shape that indicates periodic motion (for small angles, where we expect non-chaotic behavior).
- **Chaotic Behavior:** In a chaotic system, the trajectory in phase space never forms a closed orbit, and predicting its behavior becomes practically impossible. On the Poincaré map, this is reflected by non-repeating points scattered in the vicinity of a region. Additionally, the chaotic nature often leads to more "open" patterns, where the points do not cluster but instead spread across a larger area of the map. As the system becomes more chaotic, these patterns grow increasingly irregular, capturing more volume and reflecting the unpredictable and sensitive nature of the dynamics. Chaotic systems also exhibit lack of periodicity and irregular spread of points on the Poincaré map. As the degree of chaos increases, the system's behavior becomes more unpredictable, and the map transitions from tight clusters to open, scattered regions, representing a more extensive occupation of phase space.

2 Experimental Setup

The experimental setup consists of a double pendulum, one rod connected to the table central axis, and another rod connected to the end of the first one by a pivot. The table is attached to the floor and wall to reduce energy loss. Motion is tracked using high-speed cameras (240 fps) and analyzed using color-based marker detection code.

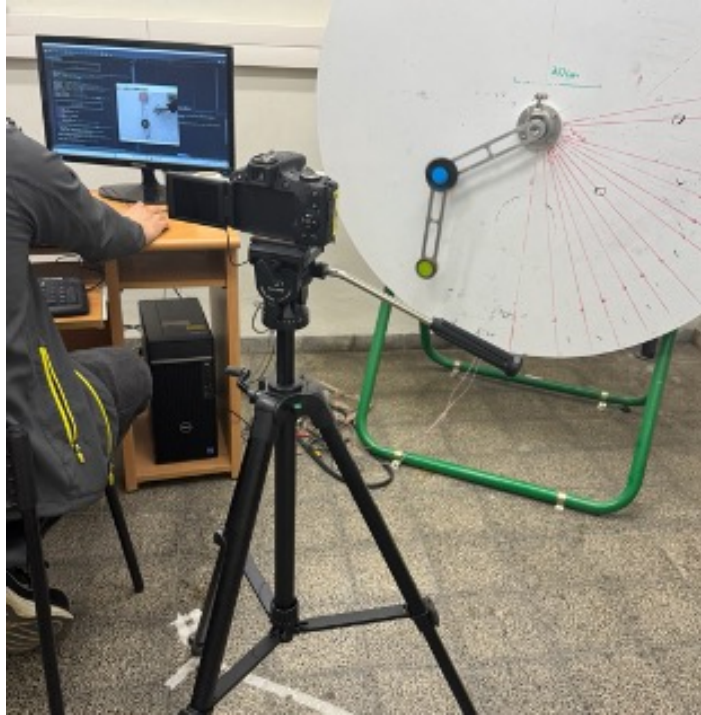


Figure 3: Experimental Setup

2.1 System's Parameters

The measured double pendulum parameters are shown in Table 1.

Parameter	Value	Uncertainty	Units
Mass of pendulum 1 (m_1)	0.436	0.001	kg
Mass of pendulum 2 (m_2)	0.497	0.001	kg
Length of pendulum 1 (l_1)	0.215	0.002	m
Length of pendulum 2 (l_2)	0.215	0.002	m
Length to center of mass (l_3)	0.255	0.002	m
Moment of inertia 1 (I_1)	0.023	0.002	kg·m ²
Moment of inertia 2 (I_2)	0.023	0.002	kg·m ²
Period of pendulum 1 (T_1)	1.1	0.1	s
Period of pendulum 2 (T_2)	1.2	0.1	s
Gravitational acceleration (g)	9.80665	0.00001	m/s ²

Table 1: System parameters and their uncertainties used for the analysis of the double pendulum.

2.2 Detection Code

The detection code is a crucial part of the experimental setup, designed to accurately track the motion of the double pendulum’s two masses from high-speed video recordings at 240 frames per second. The process begins with an interactive calibration step, where the user manually selects a reference point (origin) on the first video frame. Subsequently, regions of interest (ROIs) are defined around each mass to extract their color profiles in the HSV (hue, saturation, value) color space. This ensures that the detection is reliable since it already adapts itself to variations in lighting conditions. For each frame, the detection algorithm applies the defined HSV thresholds to isolate pixels corresponding to the colors of the two masses. The resulting binary masks are processed to extract contours, and the centroid of the largest contour in each mask is calculated to represent the position of the respective mass. In cases where a mass is not detected due to factors such as occlusion or motion blur, the code interpolates the missing positions linearly using the nearest detected points so it ensures the continuity of the data. The tracked positions of the masses, recorded relative to the selected origin, are saved along with a time array, in CSV files for further analysis. Additionally, the code generates us a validation video where the detected positions are visually marked on each frame, serving us as a visual aid for validating the detection process.

3 Results and Analysis

3.1 Energy Conservation

In our experiment, we assumed energy conservation as a foundational principle. This assumption simplifies the EOMs into an analytic framework, and enable precise calculations. Without energy conservation, friction introduces complexities that make the system's precise behavior at any given moment difficult to determine.

Friction leads to energy dissipation, causing the system's dynamics to converge toward a specific state. Systems with higher energy conservation exhibit less chaotic behavior. Consequently, identifying the temporal regime where energy conservation remains valid is critical for meaningful analysis and reliable conclusions.

To quantify energy conservation, we analyzed the system's Hamiltonian $H(t)$ as a function of time. Utilizing the experimental coordinates of $\varphi(t)$ and $\dot{\varphi}(t)$, we could substitute them into the Hamiltonian (energy) expression $H = T + V$, according to Equations (2) and (3). By plotting $H(t)$, we observed a decay due to friction. To characterize this decay, we fitted an exponential function to the data:

$$H(t) = H_0 e^{-t/\tau},$$

where H_0 is the initial energy, and τ is the decay constant representing the system's dissipation timescale.

From the fit, we determined $\tau_{\min}$, the smallest decay constant across all initial conditions. This provides an upper bound for the time within which the system's behavior can be approximated as an energy-conserving one. Specifically, for $t < \tau_{\min}$, the system can be treated analytically as nearly conserving energy. Below is shown the experimental data along with an exponential fit of the system's energy that provided us the minimal decay time constant $\tau_{\min} \approx 30$ sec.

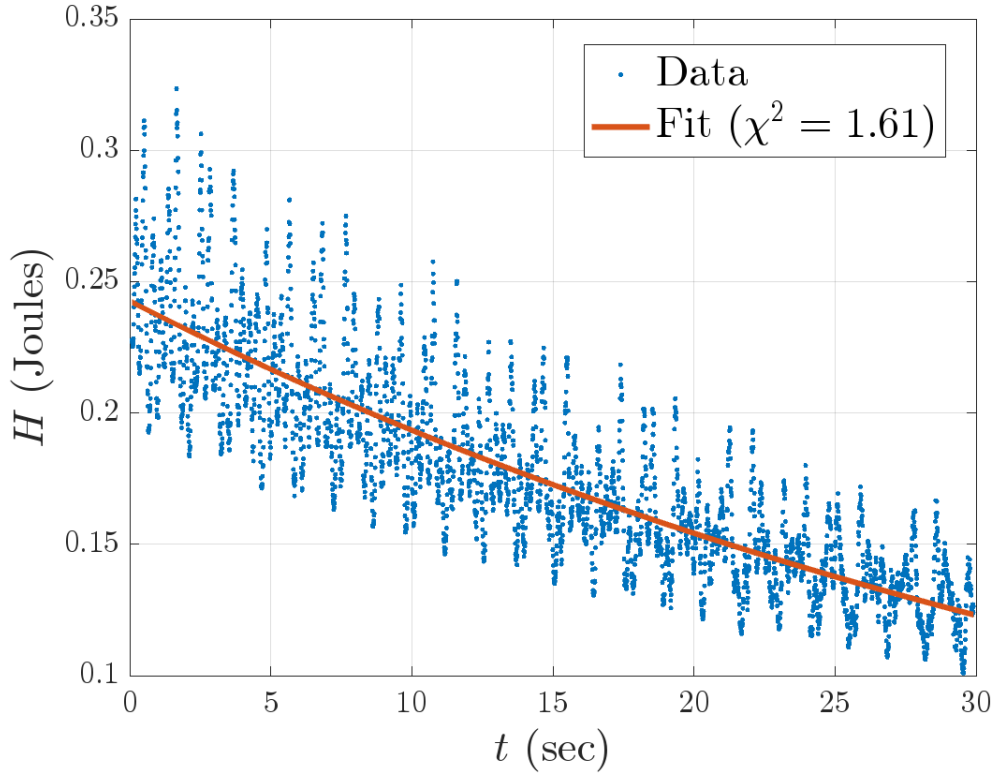
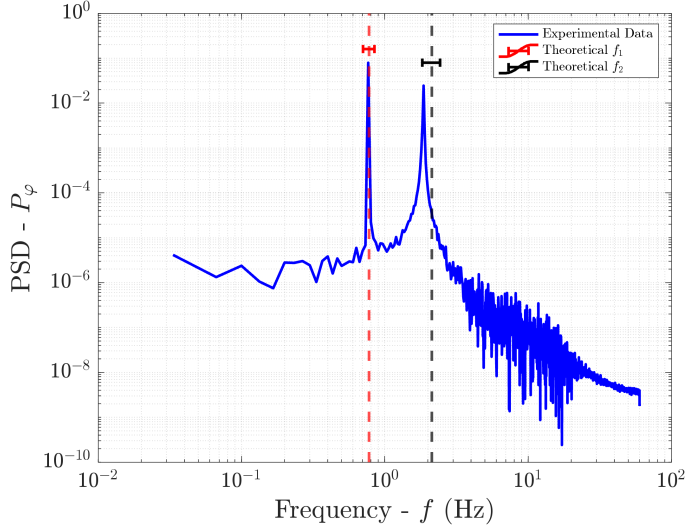


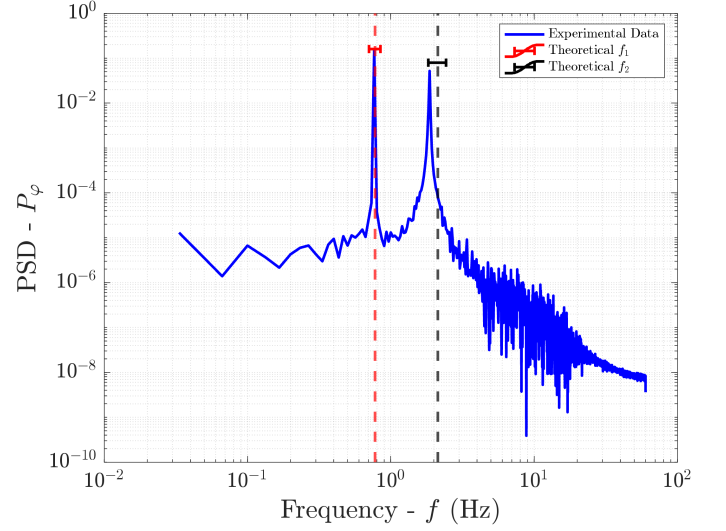
Figure 4: Temporal evolution of the system's Hamiltonian $H(t)$. The experimental data is represented by discrete points, while the solid line shows the fitted exponential decay $H(t) = H_0 e^{-t/\tau}$. The decay constant $\tau_{\min} \approx 30$ sec was determined, providing an estimate for the system's dissipation timescale and the validity regime of energy conservation.

3.2 Fourier Analysis for Small Angles Approximation

The power spectral density for both angular displacements is plotted in Figures 5a and 5b. The theoretical frequencies are computed using Equation (11), and marked on the plots with error bars to show agreement with experimental data.



(a) Fourier spectrum of $\varphi_1(t)$ showing dominant experimental and theoretical frequencies.



(b) Fourier spectrum of $\varphi_2(t)$ showing dominant experimental and theoretical frequencies.

Figure 5: Fourier spectra of the angular displacements $\varphi_1(t)$ and $\varphi_2(t)$. Power spectral density (PSD) P_φ is shown against frequencies f spectrum, all axes are logarithmic. The theoretical frequencies are highlighted and compared with experimental results.

One can clearly denote two distinct resonance peaks. These align well with the theoretical computed frequencies:

$$\begin{aligned} f_{\text{theo},1} &= 0.77 \pm 0.04 \text{ Hz} \\ f_{\text{theo},2} &= 1.87 \pm 0.04 \text{ Hz} \end{aligned}$$

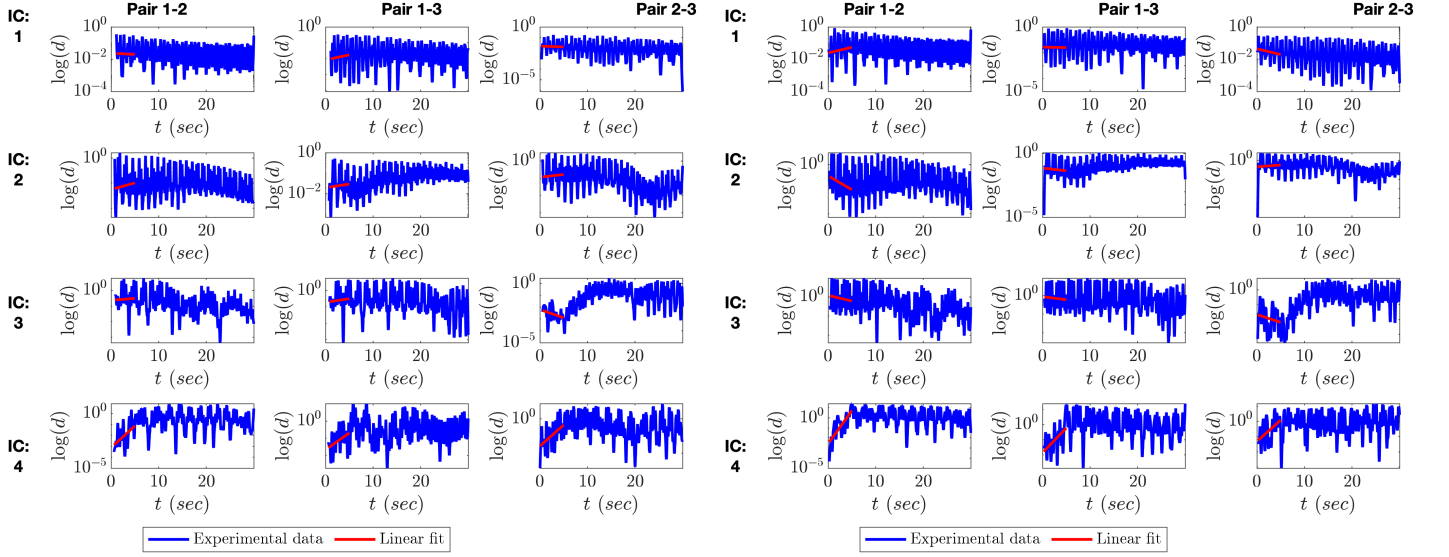
where the confidence intervals of these include the experimental frequencies:

$$\begin{aligned} f_{\text{exp},1} &= 0.7 \pm 0.07 \text{ Hz} \\ f_{\text{exp},2} &= 2.2 \pm 0.03 \text{ Hz} \end{aligned}$$

This analysis demonstrates that the small-angle approximation of the double pendulum's motion yields two distinct frequency modes, corresponding to the symmetric and antisymmetric oscillations. The experimental results align well with the theoretical predictions, validating the small angles approximation model within the limits of uncertainty.

3.3 Lyapunov Exponents

Using Equation (16), we could calculate the distance between the trajectory measured in one experiment and its value in a different experiment, trying to start the experiment with an identical initial condition $\varphi_1(0)$, although not possible experimentally. Obviously, without this accuracy limit in experimental initial condition, the trajectories would not separate since the double-pendulum system is deterministic. It is worth noting that the inaccuracy in the numerical simulations of Section 1.3, which allows chaotic behavior, arises from the numerical computer storage accuracy. Herein we show the experimental distances logarithm for each pair of experiments.



(a) Logarithm of filtered experimental distances for φ_1 over time. (b) Logarithm of filtered experimental distances for φ_2 over time.

Figure 6: Experimental trajectories separation logarithm $\log(d)$ for φ_1 and φ_2 , along with linear fit applied until $t = 5$ sec. The trajectory distances d are computed for three experiments pairs (1-2, 1-3, 2-3), and for three desired initial conditions: 1 - $\varphi_1^0 = 15^\circ \pm 3^\circ$, 2 - $\varphi_1^0 = 54^\circ \pm 3^\circ$, 3 - $\varphi_1^0 = 83^\circ \pm 3^\circ$, 4 - $\varphi_1^0 = 102^\circ \pm 3^\circ$; where the error was approximated to our ability to determine visually the first arm angle $\approx 3^\circ$.

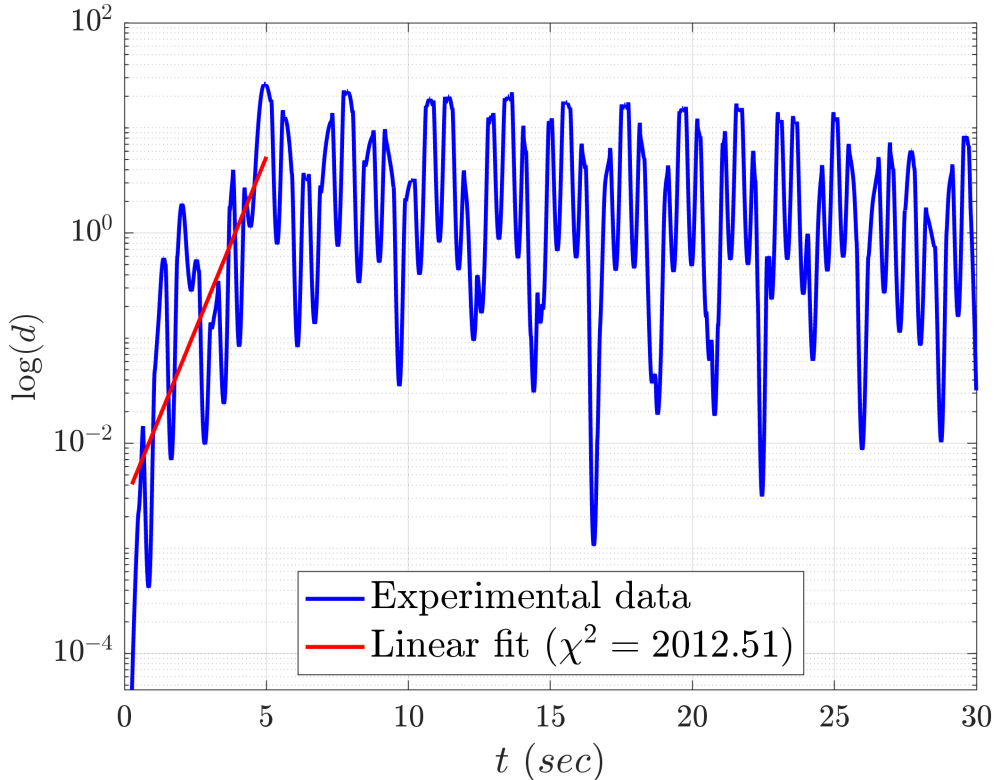


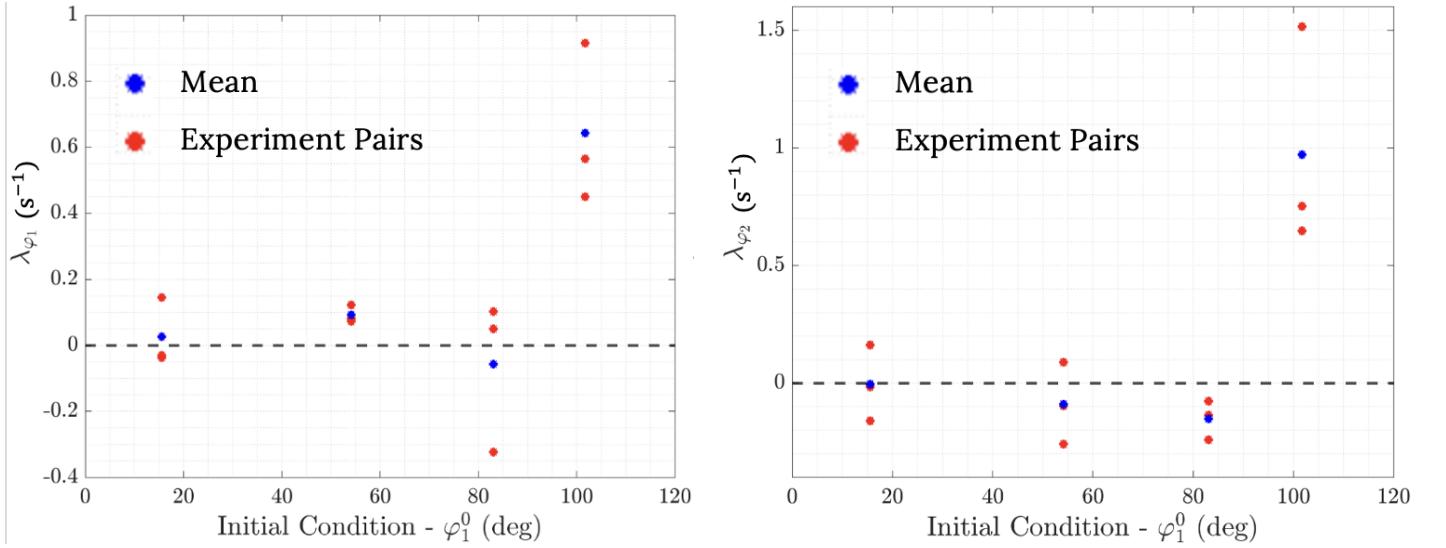
Figure 7: Experimental trajectory separation along with the linear fit which exhibits the largest slope (Lyapunov exponent) observed. This appears to happen for φ_2 , experimental pair 1-2, and the largest initial condition $\varphi_1^0 = 102^\circ \pm 3^\circ$.

It is worth noting that the region chosen for applying linear fits is not linear (can be seen by the large value of χ^2), however, it represent the growth region for the trajectories separations, therefore still incorporates information about the system's chaotic/converging behavior. Computing the mean values of the slopes for all the pair combinations, we compute the Lyapunov exponent for each initial condition; obviously this value would be accurate only for an infinite number of experiments. The Lyapunov exponents for each initial condition and initial condition, these are summarized in the table below:

$\varphi_1^0(^{\circ})$	$16 \pm 3^{\circ}$	$54 \pm 3^{\circ}$	$83 \pm 3^{\circ}$	$102 \pm 3^{\circ}$
$\lambda_{\varphi_1}(s^{-1})$	0.026 ± 0.1	0.092 ± 0.03	-0.057 ± 0.2	0.643 ± 0.2
$\lambda_{\varphi_2}(s^{-1})$	-0.005 ± 0.2	-0.089 ± 0.2	-0.151 ± 0.08	0.971 ± 0.5

Table 2: Lyapunov Exponents λ_{φ_1} and λ_{φ_2} computed averaging over experiment pairs fit slopes. The error values are computed using the appropriate standard deviations.

From these results we extract two important values that characterize the chaotic behavior of the system, as explained in Section 1.3, which are the largest Lyapunov exponent for each angles - $\lambda_{\varphi_1} = 0.643 \pm 0.2$ and $\lambda_{\varphi_2} = 0.971 \pm 0.5$. Both values are positive in all their confidence intervals, and approve a chaotic behavior at high initial conditions. Herein we show the computed Lyapunov exponents for each pendulum angle against the initial conditions:



(a) Lyapunov exponent λ_{φ_1} for the first pendulum angle as a function of initial conditions φ_1^0 . (b) Lyapunov exponent λ_{φ_2} for the second pendulum angle as a function of initial conditions φ_1^0 .

Figure 8: Computed Lyapunov exponents for each pendulum angle (λ_{φ_1} and λ_{φ_2}) plotted against the initial conditions.

These graphs were created to visualize the transition to chaos via our experimental results, similarly as the numerical values visualized for different system parameters in Figure 2. In a similar way, we observe varying values around zero, negative and positive, for initial conditions below $\varphi_1^0 \approx 83^{\circ}$. At $\varphi_1^0 \approx 102^{\circ}$, the values become positive clearly positive, as already pointed out above. This plots allow us to determine an approximation of the region at which the system transits from a converging behavior to a chaotic one: $\varphi_1^0 \in [83, 102]^{\circ}$. Comparing this region to the numerical one observed in Figure 2 would not be meaningful since the system's parameters are different, although our results follow a similar behavior which provides us a small confirmation of the validity of our analysis in this section.

3.4 Poincaré Maps

As explained in Section 1.4, the Poincaré maps reduce the dimensionality of the system, allowing us to identify periodicity or chaos through the distribution of points in the phase space.

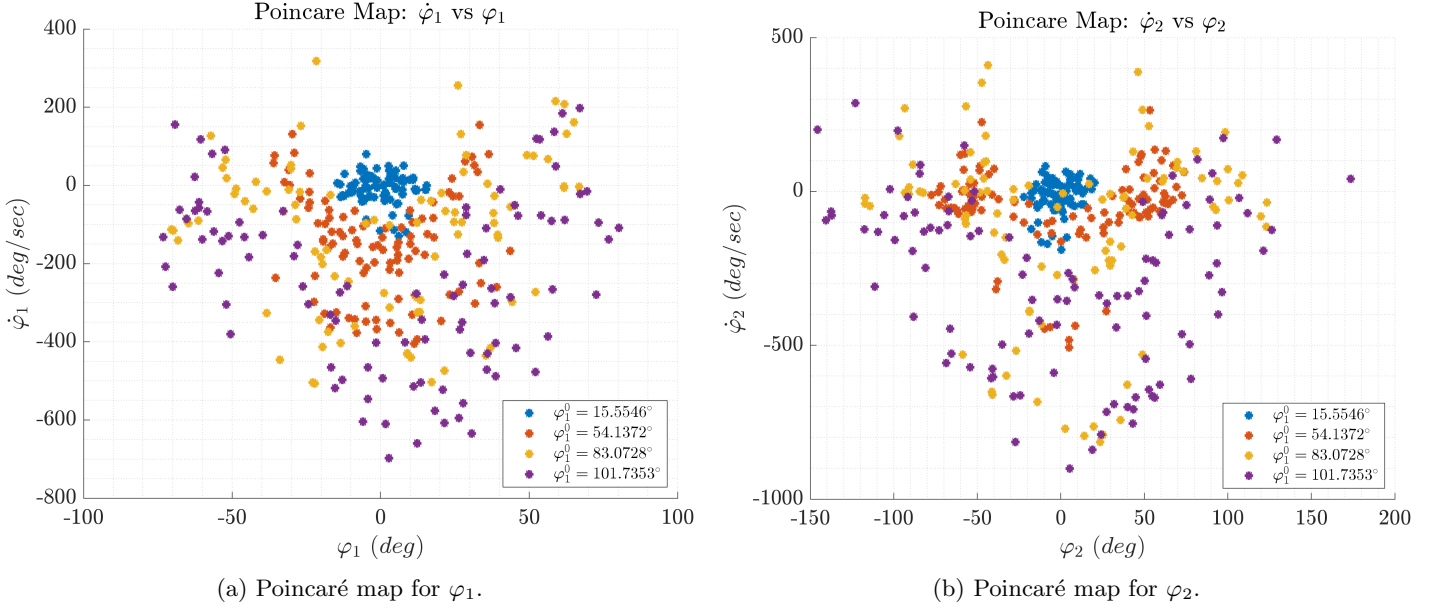


Figure 9: Poincaré maps for the double pendulum.

The Poincaré maps for the angular displacements φ_1 and φ_2 are presented in Figure 9. For small initial angles, while for the initial conditions verified - this applies only for $\varphi_1^0 = 16^\circ \pm 3^\circ$, we observe closed, well-defined loops, consistent with periodic motion. This agrees with what has been said in Section 1.4, where small angle dynamics were shown to approximate linear behavior. As the initial angles increase, the patterns on the Poincaré map become scattered and non-repetitive. This transition marks the starting transition from a periodic to a chaotic behavior and is characterized by irregular spacing and non-overlapping points (non-periodicity in phase-space). The chaotic behavior is consistent with the positive Lyapunov exponents observed in Section 3.3, reinforcing the sensitivity of the system to initial conditions.

4 Discussion

The experimental results effectively demonstrate the transition from periodic to chaotic behavior in the double pendulum system. The analysis of energy conservation, Fourier, Lyapunov exponents, and Poincaré maps collectively gives us a comprehensive picture of the dynamics of the system. Energy conservation, as analyzed in Section 3, shows an exponential decay which we linked to friction, and provide us a timescale for when the system can be approximated as conservative. The Fourier analysis in Section 3.2 validate the theoretical small-angle approximations, which gave two dominant modes - symmetric and antisymmetric, since we observed two distinct resonance peaks (of the experimental data) which are included inside the confidence interval of the theoretical frequencies. The Lyapunov exponents calculated in Section 3.3 quantify the chaotic nature of the system at large initial conditions, with a transition estimated within the range $\varphi_1^0 \in [83, 102]^\circ$. The positive values confirm the exponential divergence of trajectories, which also confirms chaotic behavior. Finally, the Poincaré maps in Section 3.4 visually represent the transition from periodic to chaotic dynamics. For small angles, the closed loops (approximately) in the maps confirm regular, predictable motion. For larger angles, the scattered points reflect the system's sensitive dependence on initial conditions and its unpredictable behavior. The neglected but still present friction and measurement inaccuracy may introduce noise, which could partially account for deviations between theoretical and experimental results. However, the overall agreement validate the analytical framework and experimental methodology we used to characterize chaos in our double pendulum system.

5 Conclusions

The double pendulum is a powerful example of how even simple mechanical systems can evolve into complex, unpredictable behavior. In this study, we explored its dynamics and saw how chaos emerges as the initial conditions change. The tools we used—like Fourier analysis, Lyapunov exponents, and Poincaré maps have proven to be valuable for identifying and understanding this transition. Lyapunov exponents, in particular, stood out as a reliable way to estimate the onset of chaos. By measuring how quickly small differences in starting conditions grow, we were able to quantify when the system moves from orderly motion to unpredictable chaos. This shows how useful these exponents are for studying nonlinear

systems. When we compared our results to previous work, like [1], it became clear that the angle φ_1^0 where chaos begins is not universal, it depends on the specific properties of the system, like the masses and lengths of the pendulum arms.. The small-angle approximation was another key takeaway. It worked well for predicting motion at low angles, but as the angles got larger, the system's nonlinear nature took over, leading to chaotic, unpredictable behavior. This demonstrates how sensitive systems like this are to changes in their starting conditions. Lastly, energy conservation played an important role in simplifying our analysis. By identifying a timescale where energy losses due to friction are minimal, we were able to make meaningful comparisons with theoretical predictions. In the end, the double pendulum is more than just a theoretical curiosity. It is a good reminder of how even deterministic systems can behave in ways that feel random and unpredictable, teaching us about the nature of chaos and the complexity of our world.

Acknowledgements

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References

- [1] R. B. Levien and S. M. Tan. Double pendulum: An experiment in chaos. *American Journal of Physics*, 61:1038–1044, 1993.